

Leopoldo Cuspinera

Senior Data Scientist

Graph Analytics — Network Intelligence — Forecasting

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SUMMARY

Senior Data Scientist with 5+ years of experience delivering graph analytics, forecasting solutions and large-scale data products for Electricity Networks. PhD-trained physicist specialising in transforming complex network data into scalable analytical capabilities through Neo4j, Spark and Databricks.

KEY ACHIEVEMENTS

- **ForeSIGHT Load Allocation Methodology:** Designed and proposed a scalable load-allocation methodology that replaced the original tender approach. Reformulated the problem using network topology, flow-conservation principles and measurement reconciliation, creating a reusable analytical capability adopted by the customer.
- **Graph Platform Ownership:** Defined the Neo4j-based network data model and graph architecture underpinning critical capabilities within the ForeSIGHT programme.
- **Technical Leadership:** Provided onboarding and technical guidance to Data Engineers and Subject Matter Experts, influencing implementation approaches, data modelling decisions and analytical methodologies.
- **Large-Scale Network Analytics:** Designed analytical solutions for electrical networks containing more than 12 million assets and multi-year measurement datasets using Neo4j, Spark and Databricks.

EXPERIENCE

- **Camlin Group**

Data Scientist

Remote, Italy

Jun 2021 – Present

- **ForeSIGHT Load Allocation Methodology:** Identified limitations in the original load-allocation approach proposed in the SPENW ForeSIGHT tender and reformulated the problem from first principles. Designed a mathematically rigorous framework based on network topology, flow-conservation principles and measurement reconciliation, creating a scalable methodology that was subsequently adopted by the customer.
- **Graph Architecture and Network Data Model:** Defined the Neo4j-based graph architecture and data models supporting physical network connectivity analysis across electrical networks containing more than 12 million nodes and 13 million relationships. Integrated network assets, measurements and connectivity into a reusable analytical foundation used across multiple network-intelligence capabilities.
- **Technical Leadership and Mentoring:** Provided technical onboarding and guidance to 4 Data Engineers and 2 Subject Matter Experts. Regularly influenced implementation approaches, analytical methodologies and data-modelling decisions while acting as a technical reference point across the project.
- **Reusable Graph Analytics Platform:** Designed and developed reusable Python tooling around Neo4j to support network traversal, connectivity analysis, missing-data estimation, load tracking and graph-based investigations. Reduced duplication of effort across analytical use cases and improved consistency of network analysis.
- **Customer Collaboration and Solution Design:** Worked directly with Distribution Network Operators and stakeholders to gather requirements, demonstrate analytical capabilities and translate operational challenges into scalable technical solutions. Contributed to shaping customer-facing functionality within ForeSIGHT and Sapient Network Insights.
- **Forecasting, Data Quality and Network Analytics:** Developed forecasting, data-quality and geospatial analytical solutions using Python, Spark and Databricks. Supported network planning, maintenance prioritisation, capacity analysis and operational decision-making through large-scale analysis of electrical-network measurements.

- **Esosphaera S.R.L.**

Data Scientist

Treviso, Italy

Sep 2020 – Jun 2021

- **Python:** Created ETL pipelines that queried different tables of the main database via Postgres, created features with pandas, updated the curated data on S3 buckets and made use of Athena to feed Tableau dashboards.
- **Tableau:** Created several interactive Dashboards that allowed both salesmen and stakeholders in understanding the evolution of the most impactful interactions between end users and customers at a glance.
- **Perimeter Institute** Waterloo, Canada
Visiting Researcher *Jan 2016 - Mar 2019*
 - **Mathematica, Python:** Made extensive use of both Mathematica and Python to solve the Partial Differential Equations that describe vacuum decays around higher dimensional black holes. Worked in an international collaboration that had as a result two of the papers I published during my PhD.
 - **Guest speaker:** Gave a seminar explaining the calculations we made to obtain the results of our work.
- **Durham University** Durham, England
Teaching Assistant *Oct 2015 - Nov 2019*
 - **Maths and stats lab:** guided students with different academic backgrounds in finding their own answers to mathematical and statistical problems. Due to the positive impact our team had on the students, we obtained the 'Student employee of the year (2019)' award.
 - **Tutor:** Helped students in understanding concepts in Quantum Mechanics, Linear Algebra, Statistics and Calculus
- **Deutsches Elektronen-Synchrotron** Hamburg, Germany
Internship *Jun 2014 - Sep 2014*
 - **C++:** Studied the predictions of different Monte Carlo simulators on diffractive dissociations of proton-proton collisions and compared with real data measured at LHC. This was my first time being exposed to C++, which demonstrates I can quickly pick up a programming language, develop a project and provide results in short time.

SKILLS

- **Programming and Data Engineering:** Python, SQL, Spark, PySpark, Databricks, Delta Lake, Neo4j, Cypher, Git, Bash, Pandas
- **Analytics and Machine Learning:** Forecasting, Time Series Analysis, Graph Analytics, Machine Learning, Statistical Modelling, Clustering, Anomaly Detection, Data Quality
- **Domain Expertise:** Electricity Networks, Network Modelling, Physical Connectivity, Distribution Network Operators (DNOs), Capacity Analysis, Load Allocation
- **Visualisation and Geospatial:** Tableau, Plotly, Dash, GeoPandas, Shapely, Folium
- **Leadership and Communication:** Technical Leadership, Stakeholder Engagement, Customer Collaboration, Mentoring, Project Ownership, Technical Problem Solving
- **Languages:** English, Spanish, Italian

EDUCATION

- **Durham University** Durham, England
PhD in Physics *Oct 2015 – Jan 2020*
- **Durham University** Durham, England
*MSc in Strings Particles and Cosmology. **Honours:** Distinction* *Oct 2014 – Sep 2015*
- **Benemerita Universidad Autonoma de Puebla** Puebla, Mexico
BSc. in Physics *Aug 2009 – Jun 2014*

PUBLICATIONS

- **Black holes, vacuum decay and thermodynamics**
PhD thesis Jan 2020
- **Are Superentropic black holes superentropic?**
Journal of High Energy Physics Nov 2019
- **Higgs vacuum decay in a braneworld**
International Journal of Modern Physics D Nov 2019
- **Higgs Vacuum Decay from Particle collisions?**
Physical Review D Jan 2019